System Design Workshop: LLM-powered Search at Scale

Scaler School of Technology | ML System Design and LLMOps | Week 6 — Session 1/2 | Student Worksheet

The ML System Design Rubric — Your Framework for Today

Use this 10-step framework to design the system. Work through the steps in order on pages 2–3.

1. **Clarify the Problem** — scope, constraints, success metrics
2. **Context** — business objective, users, what failure costs
3. **Problem Statement** — input, output, what to predict/rank
4. **Define Requirements** — functional and non-functional
5. **Estimations** — users, QPS, data volume, storage, compute
6. **System Architecture** — (a) Data Collection • (b) Data Engineering • (c) Feature Engineering • (d) Preprocessing • (e) Models • (f) Post-processing
7. **Evaluation** — offline metrics, online north-star, guardrails
8. **A/B Testing** — unit, split, duration, biases to flag
9. **Deployment & Monitoring** — pipelines, roll-out, dashboards
10. **Risk & Trade-offs** — latency vs. accuracy, ethical risks, failure modes

The Problem

Design a semantic search system for a large content platform (e.g., Stack Overflow, GitHub, enterprise knowledge base) with **100 million documents**.

Requirements: sub-500ms p99 latency • 5,000 QPS • LLM-augmented answer summaries • hybrid retrieval (dense + sparse BM25) • new documents indexed within **1 hour** of ingestion

Steps 1–3 — Clarify, Context, Problem Statement

Ask your clarifying questions, note the context, then write a precise ML problem statement.

Clarifying questions I would ask (think: what is a "result"? when is LLM needed?):

Context (business objective, users, cost of failure):

Problem statement (input, output, constraint):

Step 4 — Requirements

Requirement	Value / assumption	Implication for design
Latency SLA		
Throughput (peak QPS)		
Availability		
Doc freshness SLA		
LLM cost per query		

Step 5 — Estimations

What	Your estimate	Design implication
Dense vector index size		
BM25 index size		
Concurrent requests		
LLM cost per day		

Step 6 — System Architecture

Label each component: what it does, key tools, key decisions. Draw the full system diagram in the box below.

(a–c) Data Collection, Engineering & Feature Store

(d–e) Hybrid Retrieval, Reranker & LLM Augmentation

(f) Post-processing, Caching & Full System Diagram

Draw components as labelled boxes. Show data flow with arrows. Label every arrow. Include the LLM cache and fallback path.

Step 7 — Evaluation

Type	Metric(s)	Why this metric?
Offline		
Online north-star		
LLM guardrail		

Step 8 — A/B Testing

Unit of randomisation, split, duration, sample size, and one bias specific to LLM-augmented search you would flag.

Step 9 — Deployment and Monitoring

Deployment (pipelines, rollout, index upgrade strategy):

6 metrics to monitor:

1.
2.
3.
4.
5.
6.

Step 10 — Risk Management and Trade-offs

Trade-off	Your choice	Reasoning
Dense vs. sparse vs. hybrid retrieval		
LLM always vs. conditional on intent		
Self-hosted vs. API embedding model		
Chunking size (small vs. large)		
Reranker (yes vs. no)		

What Breaks First When the Embedding Model is Upgraded?

Describe the failure mode, how to detect it silently, and how you would safely deploy an embedding model upgrade.

Name: _____ Group: _____ Date: _____